## DATA INSPECTION AND PREPARATION

The Global Superstore Orders dataset was inspected and prepared in Power Query before being used for dashboard development. The preparation process focused on data structure, field types, record integrity and text consistency.

The Orders table contained **51,290 rows and 24 columns**. The following checks and transformations were completed:

- **Data type validation:** Reviewed column data types and ensured that Order Date and Ship Date were recognised as date fields. Numeric fields, including Sales, Profit, Quantity, Discount and Shipping Cost, were assigned appropriate data types.
- **Text standardisation:** Applied Trim and Clean transformations to selected text columns to improve consistency and reduce issues caused by unwanted spaces or non-printing characters.
- **Record integrity:** Checked Row ID for duplicate and blank values. The table retained 51,290 rows after these checks.
- **Order identification:** Recognised that Order ID can appear on multiple rows because an order may contain several product line items. Consequently, order volume is measured using distinct Order IDs rather than counting transaction rows.
- **Missing data assessment:** Identified substantial missingness in Postal Code, with approximately 81% of values missing in the initial data profile. Postal Code was treated as a text field, and missing values were retained rather than replaced with fabricated information.
- **Record preservation:** Retained the original transaction records while carrying out the preparation steps.

These checks improve the consistency of the dataset used for analysis. However, the missing Postal Code values remain a documented limitation, and further validation of analytical results is necessary before management recommendations are finalised.

## KEY PERFORMANCE INDICATORS AND DEFINITIONS

The dashboard uses a defined set of key performance indicators (KPIs) to assess sales performance, profitability and order value. Consistent definitions allow users to interpret results across reporting periods, product categories, regions and customer segments.

| KPI | Definition | Business purpose |
|---|---|---|
| Total Sales | Sum of all recorded sales values | Measures the overall sales generated during the selected period. |
| Total Profit | Sum of all recorded profit values | Measures the total profit recorded in the dataset. |
| Total Orders | Distinct count of Order ID | Measures the number of unique orders rather than individual product line items. |
| Average Order Value | Total Sales divided by Total Orders | Measures the average sales value per unique order. |
| Profit Margin | Total Profit divided by Total Sales | Measures profit relative to sales and helps assess profitability alongside revenue. |

These KPIs are intended to be interpreted together. High sales do not necessarily indicate strong profitability, while a high profit margin does not, on its own, indicate a large contribution to total profit. Comparing revenue, profit, order volume and margin provides a more balanced view of commercial performance.

The KPI values presented in the final report will be validated against the Power BI dashboard before the findings and recommendations are confirmed.

## DASHBOARD STRUCTURE AND MANAGEMENT QUESTIONS

The Power BI dashboard is organised into three analytical pages. Each page addresses a distinct set of management questions and supports further investigation through relevant visualisations and interactive filters.

| Dashboard page | Primary business questions | Intended management use |
|---|---|---|
| Executive Performance | How are overall sales, profit, order volume and profit margin performing? How have sales and profit changed over time? How do categories, regions and customer segments contribute to sales? | Monitor overall commercial performance, identify significant trends and determine which areas require further investigation. |

| | | |
|---|---|---|
| Profitability and Product Intelligence | Which product categories and sub-categories generate the strongest profit? Which products have low or negative profit? How do sales, profit and discount levels compare? | Identify products and sub-categories for further profitability review, investigate weak results and inform product-level decisions. |
| Regional and Customer Intelligence | How do sales and profit vary across geographic markets and customer segments? Where is performance concentrated? | Compare market and customer performance, identify areas for further analysis and support more targeted commercial planning. |

Interactive filters allow users to examine results across relevant dimensions. The dashboard is intended to help management move from a high-level performance observation to a more focused investigation of the products, markets, customers or periods associated with that result.

The dashboard identifies patterns and areas requiring attention; it does not automatically establish the causes of those patterns. Management actions should therefore be based on verified findings, appropriate business context and further investigation where necessary.